The empirical recurrence for OEIS A234545: recovering a conjecture that is not written down

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Abstract

OEIS A234545 records "Empirical recurrence of order 49", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1296$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 49, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 49 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234545 is "Number of (n+1) X (3+1) 0..5 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 3 (constant-stress 1 X 1 tilings).". It has offset 1 and begins

$$7056, 39188, 217304, 1207448, 6716768, 37416120, 208681224, 1165446648, \ \ldots$$

The entry states:

> Empirical recurrence of order 49 (see link above).

As of the "Last modified" line on the live entry (Jun 20 2022 19:15:49, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1296$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S =$ 1296.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2592$ exact terms of the model returns order 49 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{49} c_i\, a(n-i),$$

c_1	26	c_{14}	−5408269056220	c_{27}	2241457073664239872	c_{40}	−3767506974081548288
c_2	−12	c_{15}	37675854476648	c_{28}	−3740653408721728256	c_{41}	−2272101201256382464
c_3	−5260	c_{16}	43640976326500	c_{29}	−4831801539456774656	c_{42}	1230992312354471936
c_4	35181	c_{17}	−587766323647328	c_{30}	7801713964264560640	c_{43}	702584852859322368
c_5	414221	c_{18}	42105246349032	c_{31}	8123976408299915264	c_{44}	−262477555716587520
c_6	−4851762	c_{19}	6105275423800768	c_{32}	−12136072455198738432	c_{45}	−142683773085941760
c_7	−12867246	c_{20}	−4877318234656832	c_{33}	−10686815328089985024	c_{46}	32289151411814400
c_8	329232692	c_{21}	−43268039721638816	c_{34}	14174822937073090560	c_{47}	16787436679987200
c_9	−233146807	c_{22}	53679733259664480	c_{35}	10954730117178720256	c_{48}	−1695975211008000
c_{10}	−13391633880	c_{23}	217288110624368320	c_{36}	−12401172724166656000	c_{49}	−847987605504000
c_{11}	36612619396	c_{24}	−326736152478112896	c_{37}	−8654673705366781952		
c_{12}	343393138802	c_{25}	−802280739297632128	c_{38}	8033505589811085312		
c_{13}	−1551214821662	c_{26}	1314108156041592064	c_{39}	5172094599294091264		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 49, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $49 + 4$ terms removed returns order 49 as well, so $\deg q_{\min} = 49 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $49 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 49 that this sequence satisfies — and that family turns out to have exactly one member.

References

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